Ordering fractions in different ways

Task A: Comparing fractions using a common numerator

1) Insert either < or > to make the statement true.

a) $\frac{13}{21} \boxed{<} \frac{13}{15}$

c) $\frac{7}{11} \boxed{<} \frac{7}{9}$

b) $\frac{8}{15} \boxed{>} \frac{8}{19}$

d) $\frac{11}{315} \boxed{<} \frac{11}{311}$

e) Place the words 'larger' and 'smaller' in the sentence:
The *larger* the denominator the *smaller* the 'parts'.

2) Write the following in order from smallest to largest.

a) $\frac{10}{21}, \frac{10}{18}, \frac{10}{35}, \frac{10}{15}$ $\qquad$ $\frac{10}{35}, \frac{10}{21}, \frac{10}{18}, \frac{10}{15}$

b) $\frac{234}{512}, \frac{234}{518}, \frac{234}{500}, \frac{234}{511}$ $\qquad$ $\frac{234}{518}, \frac{234}{512}, \frac{234}{511}, \frac{234}{500}$

3a) Place the words 'larger' and 'smaller' in the sentence:

$\frac{3}{11}$ is *larger* than $\frac{3}{12}$ but *smaller* than $\frac{3}{10}$.

b) Jacob says '$\frac{3}{11}$ is exactly halfway between $\frac{3}{10}$ and $\frac{3}{12}$.'
Is he correct?

Jacob is not correct, it is closer to $\frac{3}{12}$

c) Find another fraction between $\frac{3}{10}$ and $\frac{3}{12}$. *e.g.* $\frac{6}{23}, \frac{9}{35}, \ldots$

d) Find a fraction that is closer to $\frac{3}{10}$. *e.g.* $\frac{17}{60}, \frac{7}{24}, \ldots$

e) Find the fraction that is exactly halfway between $\frac{3}{10}$ and $\frac{3}{12}$. $\frac{11}{40}$

Task B: Ordering by comparing to one

1) Choose the most appropriate method to decide which of the following statements are true.

a) $\frac{23}{25} < \frac{25}{27}$ *True*

b) $\frac{23}{32} > \frac{23}{31}$ *False*

c) $\frac{7}{21} > \frac{258}{771}$ *False*

d) $\frac{9}{13} > \frac{7}{11}$ *True*

e) $\frac{123}{321} < \frac{123}{231}$ *True*

f) $\frac{8}{32} > \frac{137}{540}$ *False*

2) Explain why Aisha is wrong.

$\frac{19}{22}$ is larger than $\frac{22}{25}$

Aisha

E.g. Although they are both 3 'parts' away from 1, the 'parts' are larger in $\frac{19}{22}$ *so it further from 1.*

3) Use these numbers to complete the following proper fractions. 10, 11, 12, 13, 14 and 15.

a) The fraction closest to one $\frac{14}{15}$

b) The fraction closest to zero $\frac{10}{15}$

c) $\frac{10}{13} < \frac{10}{\square}$ *11 or 12*

d) $\frac{12}{15} < \frac{\square}{15}$ *13 or 14*

e) $\frac{10}{11} < \frac{13}{\square}$ *14 or 15*

f) $\frac{12}{13} > \frac{\boxed{10}}{11}$